

HOPETOUN WA, Australia

WMO: 956350

Lat: 33.931S

Lon: 120.128E

Elev: 89

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.7	42.7	30.4	24.6	64.9	34.0	28.8	57.6	28.9	56.6	25.6	57.6	7.3	40	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.6	88.4	64.3	82.5	63.8	78.0	63.7	69.6	77.0	68.3	74.6	67.3	73.1	14.7	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	102.2	71.0	66.4	97.6	70.2	65.3	93.8	69.7	33.7	77.1	32.7	74.7	31.8	73.1	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	22.9	20.6	DB	36.4	101.6	2.1	4.1	34.9	104.6	33.7	107.0	32.5	109.3	31.0	112.3	(4)
(5)			WB	35.2	73.1	2.0	1.9	33.8	74.4	32.6	75.5	31.5	76.5	30.0	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	66.6	67.1	65.9	63.6	59.7	56.2	54.7	55.3	56.7	59.3	62.3	64.2	(6)	
	DBStd	6.09	3.94	4.17	4.77	4.76	4.38	3.55	3.51	3.86	4.28	4.83	4.64	4.11	(7)	
	HDD50	9	0	0	0	0	0	1	4	2	2	0	0	0	(8)	
	HDD65	1833	22	18	41	80	175	264	320	302	251	193	107	60	(9)	
	CDD50	3995	513	477	494	408	300	187	149	167	204	287	369	440	(10)	
	CDD65	345	71	75	70	38	10	1	0	1	3	15	26	35	(11)	
	CDH74	2079	258	286	402	269	99	4	1	30	72	212	236	210	(12)	
	CDH80	785	97	103	173	89	18	0	0	6	19	90	102	88	(13)	
(14) Wind	WSAvg	10.8	12.1	11.9	11.2	9.8	9.3	9.8	10.2	10.2	11.0	11.1	11.9	11.7	(14)	
(15) Precipitation	PrecAvg	20.4	1.1	0.9	1.1	1.6	2.7	2.4	2.5	2.4	2.1	1.7	1.3	0.7	(15)	
	PrecMax	30.1	6.3	2.7	4.3	4.3	13.0	5.7	6.0	4.9	4.4	6.7	5.9	3.1	(16)	
	PrecMin	11.1	0.0	0.0	0.0	0.2	0.3	0.7	0.4	0.6	0.2	0.0	0.0	0.0	(17)	
	PrecStd	3.9	1.3	0.7	0.9	1.0	2.2	1.3	1.1	1.1	1.1	1.2	1.1	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.9	93.2	95.7	88.5	82.5	73.7	71.6	77.7	82.6	91.9	92.0	90.7	(19)	
		MCWB	67.4	68.7	67.3	62.9	61.0	56.6	55.4	56.7	58.1	60.8	62.2	64.2	(20)	
	2%	DB	80.2	81.1	84.1	81.5	76.3	69.7	67.9	71.1	74.4	79.7	79.9	79.1	(21)	
		MCWB	67.7	66.7	66.1	62.7	59.9	55.5	55.3	56.4	58.0	60.2	62.2	65.5	(22)	
	5%	DB	75.3	76.2	77.0	75.8	71.9	66.9	64.9	67.1	69.8	71.9	73.4	73.8	(23)	
		MCWB	66.4	66.7	65.3	62.2	59.2	55.9	54.8	55.8	57.1	59.7	62.6	64.4	(24)	
	10%	DB	73.0	73.6	73.3	71.9	68.4	64.4	62.7	64.0	66.1	67.5	69.8	71.4	(25)	
		MCWB	65.0	65.6	64.9	62.8	58.5	55.5	53.8	54.9	56.1	58.8	61.5	63.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.1	71.7	70.7	68.1	65.6	62.3	59.9	61.1	62.4	64.8	67.3	69.2	(27)	
		MCDB	81.6	79.3	77.7	74.4	73.9	66.2	66.0	69.4	74.2	73.7	76.1	79.2	(28)	
	2%	WB	69.3	69.5	68.8	66.5	63.0	59.5	57.6	59.1	60.1	62.8	65.4	66.9	(29)	
		MCDB	76.6	75.7	75.3	72.2	69.5	64.8	63.5	66.1	68.9	71.7	73.2	74.7	(30)	
	5%	WB	67.7	68.1	67.5	65.3	61.6	58.0	56.0	57.4	58.5	61.2	63.8	65.5	(31)	
		MCDB	73.3	73.7	73.4	70.8	67.9	63.6	62.0	64.0	66.5	69.2	70.6	72.2	(32)	
	10%	WB	66.4	66.9	66.5	64.1	60.1	56.5	54.8	55.9	57.1	59.7	62.5	64.2	(33)	
		MCDB	71.7	72.1	72.0	69.7	66.5	62.3	60.7	62.4	64.8	66.9	68.7	70.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	13.6	15.6	17.0	17.3	16.5	15.7	17.2	18.3	17.0	15.4	14.1	(35)	
		MCDBR	19.5	21.1	26.2	25.5	24.0	21.1	20.0	23.6	26.3	28.4	25.4	21.2	(36)	
	5% WB	MCWBR	10.0	10.6	11.6	11.9	11.8	11.5	10.9	11.9	12.6	12.8	11.5	10.8	(37)	
		MCWBR	16.3	16.1	17.7	17.8	18.1	16.5	16.4	19.1	21.9	21.4	19.0	18.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.357	0.352	0.336	0.323	0.308	0.299	0.295	0.305	0.326	0.334	0.340	0.343	(40)	
		taud	2.546	2.573	2.606	2.638	2.639	2.638	2.651	2.612	2.537	2.542	2.550	2.563	(41)	
	Ebn at Noon		311	306	300	287	275	269	277	288	298	308	314	316	(42)	
		Edh at Noon	35	32	29	25	22	21	22	25	31	33	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2320	2025	1675	1253	935	788	850	1131	1513	1909	2191	2396	(44)	
	RadStd		164	136	108	72	59	33	42	44	54	105	128	142	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+79	

Nomenclature: See separate page